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DEPARTMENT:-

BS-CYBER SECURITY-FALL-24 (A)

SUBJECT:-

PROGRAMMING FUNDAMENTALS (Lab)

PRESENTED BY

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PROVIDED TO

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Lab 6Lab 6A – Task 1**Code:**

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     // Declaring variables
6     bool trueValue, falseValue, grtr, grtr_eq1, lss_eq1, not_eq1 ;
7     int x , y ;
8
9     // Taking Numbers from user
10    cout << "Enter 1st number " << endl ;
11    cin >> x ;
12    cout << "Enter 2nd number " << endl ;
13    cin >> y ;
14
15    // calculating the results
16    trueValue = x < y;
17    falseValue = y == x;
18    grtr = x > y ;
19    grtr_eq1 = x >= y ;
20    lss_eq1 = x <= y ;
21    not_eq1 = x != y ;
22
23    cout << endl << "The True Value is " << trueValue << endl;
24    cout << "The False Value is " << falseValue << endl;
25
26    cout << endl << "The value of " << x << " > " << y << " = " << grtr << endl ;
27    cout << "The value of " << x << " >= " << y << " = " << grtr_eq1 << endl ;
28    cout << "The value of " << x << " <= " << y << " = " << lss_eq1 << endl ;
29    cout << "The value of " << x << " != " << y << " = " << not_eq1 << endl ;
30
31    return 0;
32 }
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 6\Lab6A-Task1.exe
Enter 1st number
7
Enter 2nd number
9

The True Value is 1
The False Value is 0

The value of 7 > 9 = 0
The value of 7 >= 9 = 0
The value of 7 <= 9 = 1
The value of 7 != 9 = 1

-----
Process exited after 4.139 seconds with return value 0
Press any key to continue . . .
```

Lab 6B – Task 1**Code:**

```
1  #include <iostream>
2  #include<iomanip>
3  using namespace std;
4
5  int main()
6  {
7      const int HIGH_SCORE = 95;
8      int score1, score2, score3;
9      double average;
10
11     // Get the three test scores.
12     cout << "Enter 3 test scores and I will average them: ";
13     cin >> score1 >> score2 >> score3;
14
15     // Calculate and display the average score.
16     average = ( score1 + score2 + score3 ) / 3.0 ;
17     cout << fixed << showpoint << setprecision ( 0 ) ;
18     cout << "Your average is " << average << endl;
19
20     // Showing the grade according to the average
21     if ( average >= 90 )
22     {
23         cout << "You got Grade A " << endl ;
24     }
25     if ( average >= 80 )
26     {
27         cout << "You got Grade B " << endl ;
28     }
29     if ( average >= 70 )
30     {
31         cout << "You got Grade C " << endl ;
32     }
33     if ( average >= 60 )
34     {
35         cout << "You got Grade D " << endl ;
36     }
37     if (average >= 50 )
38     {
39         cout << "You got Grade E " << endl ;
40     }
41     else
42     {
43         cout << "Failed ! Try hard " << endl ;
44     }
45     // If the average is a high score, congratulate the user.
46     if (average > HIGH_SCORE)
47         cout << "Congratulations! That's a high score!\n";
48
49     return 0;
50 }
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 6\Lab6B - Task1.exe
Enter 3 test scores and I will average them: 40
45
34
Your average is 40.
Failed ! Try hard

-----
Process exited after 7.615 seconds with return value 0
Press any key to continue . . .
```

// Showing the grade according to the average

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 6\Lab6B - Task1.exe
Enter 3 test scores and I will average them: 98
99
97
Your average is 98.
You got Grade A
Congratulations! That's a high score!

-----
Process exited after 5.427 seconds with return value 0
Press any key to continue . . .
```

Lab 6B – Task2**Code:**

```
1  #include <iostream>
2  using namespace std;
3  int main( )
4  {
5
6      int num1, num2;
7
8      // Taking numbnrs from user
9      cout << "Enter 1st integer " << endl ;
10     cin >> num1;
11     cout << "Enter 2nd integer " << endl ;
12     cin >> num2 ;
13     cout << "num1 = " << num1 << " and num2 = " << num2 << endl;
14
15     // Handling if numbers are not equal
16     if ( num1 != num2)
17     {
18         cout << endl << "The numbers are not equal " << endl ;
19     }
20
21     // Rprinting output if numbers are equal
22     if (num1 == num2)
23     {
24         cout<<"The values are same"<<endl;
25         cout<<"Hey that's the coincident"<<endl;
26     }
27
28     return 0;
29 }
30
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 6\Lab6B - Task2.exe
Enter 1st integer
4
Enter 2nd integer
4
num1 = 4 and num2 = 4
The values are same
Hey that's the coincident

-----
Process exited after 4.336 seconds with return value 0
Press any key to continue . . .
```

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 6\Lab6B - Task2.exe
Enter 1st integer
4
Enter 2nd integer
2
num1 = 4 and num2 = 2

The numbers are not equal

-----
Process exited after 3.995 seconds with return value 0
Press any key to continue . . .
```

Thank You for Your Time!

THE END